

TECHNICAL REPORT: SYNTHESIS OF AUTAXYS WITH NON-ARCHIMEDAN AND PROCESS ONTOLOGIES

Author: Rowan Brad Quni-Gudzinis

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Abstract

The foundational principle of **autaxys**—intrinsic self-ordering, self-arranging, and self-generating patterned existence—was introduced in 2025 as a generative meta-framework capable of explaining the ultimate origin of order, physical laws, and complex reality without recourse to external agents or pre-imposed rules. Subsequent research in 2026 has developed three concrete mathematical and computational ontologies that provide precise instantiations of autaxys’ generative engine: (1) **graph-based (Bruhat-Tits, ultrametric, p-adic) ontologies**, which give a geometric picture of hierarchical state spaces; (2) **ratio-based valuation theory**, which replaces prime-based valuations with dimensionless scaling ratios as fundamental operational parameters; and (3) **token calculus (process algebra)**, which supplies the syntactic rules for the generation of primitive events (“tokens”) from a continuous generative verb.

This report synthesizes these four lines of inquiry, demonstrating that the later ontologies are natural, concrete manifestations of the autaxic generative engine. We show how the Bruhat-Tits tree realises the autaxic dynamics of relational processing and critical state transitions; how ratio-based valuations implement the meta-logical principle of parsimony in generative mechanisms; and how token calculus formalises the precipitation of discrete nouns from a continuous Hamiltonian verb. The synthesis yields a unified **autaxic-non-Archimedean quantum-computing paradigm** that offers a path beyond the thermodynamic wall by replacing active error correction with passive ultrametric relaxation. It also provides a new foundation for explainable AI, where hierarchical data are natively embedded in p-adic spaces, and for fundamental physics, where dimensionless scaling ratios emerge as the intrinsic parameters of the autaxic generative engine.

1. Autaxys as the Generative Meta-Framework

1.1 The Generative Engine and Operational Dynamics

Autaxys is defined as “the fundamental principle of reality conceived as a self-ordering, self-arranging, and self-generating system” (Quni, 2025a). Its **generative engine** comprises five core operational dynamics (Relational Processing, Spontaneous Symmetry Breaking, Feedback Dynamics, Resonance/Coherence Establishment, and Critical State Transitions) guided by five intrinsic meta-logical principles (Intrinsic Coherence, Conservation of Distinguishability, Parsimony in Generative Mechanisms, Intrinsic Determinacy/Emergent Probabilism, and Interactive Complexity Maximization). Together, these dynamics and principles describe how patterns emerge, stabilise, and evolve into the structured phenomena we recognise as matter, energy, spacetime, and physical laws.

1.2 Meta-Logical Principles as Constraints on Generation

The meta-logical principles act as the inherent “grammar” of the generative engine. For example, the **Principle of Intrinsic Coherence** ensures that only self-consistent patterns persist; the

Principle of Parsimony favours a minimal set of generative rules that can produce maximal diversity; and the **Principle of Interactive Complexity Maximization** drives the system toward richer, stable configurations. These principles are not external laws but expressions of the intrinsic nature of autaxys itself.

1.3 Autaxys as the Foundation for Mathematical Structures

Autaxys does not prescribe a particular mathematical representation; rather, it is the ontological ground from which various mathematical structures can arise as **patterned manifestations**. The 2026 research on graph-based, ratio-based, and token-based ontologies can be viewed as different mathematical instantiations of the same autaxic generative engine. In the following sections we make this correspondence explicit.

2. Graph-Based Ontologies: Bruhat-Tits Trees as Geometric Manifestations

2.1 The Bruhat-Tits Tree as an Ultrametric State Space

Recent work on p-adic physics (Quni-Gudzinis, 2026a) models quantum state spaces as the **Bruhat-Tits tree** (T_p) , an infinite regular graph where each vertex represents an equivalence class of p-adic lattices and edges encode hierarchical adjacency. The tree’s ultrametric geometry—satisfying the strong triangle inequality $(|x+y|_p \leq \max(|x|_p, |y|_p))$ —naturally implements the **Principle of Intrinsic Coherence**: two points are either close (within the same branch) or far apart (on different branches), with no intermediate distances. This eliminates the possibility of gradual error accumulation, providing **passive geometric fault tolerance**.

2.2 Mapping Autaxic Dynamics to Tree Dynamics

- **Relational Processing** corresponds to the creation and propagation of distinctions along edges of the tree; persistent “things” emerge as stabilised sub-graphs.
- **Spontaneous Symmetry Breaking** is realised as the branching of the tree: a symmetric parent vertex breaks into (p) distinct child vertices, generating specificity.
- **Critical State Transitions** occur when a walker crosses a large hierarchical distance, jumping from one major branch to another—a discrete leap that mirrors a phase transition.
- **Feedback Dynamics** (amplification/damping) are modelled by biased random walks that preferentially stabilise certain branches, analogous to the selection of stable meta-patterns.

2.3 Applications in Quantum Memory and AI

The ultrametric geometry of the Bruhat-Tits tree has been leveraged in two concrete applications:

- **Ultrametric Relaxation Dynamics in Topological Quantum Memory** (Quni-Gudzinis, 2026b) shows that a Hamiltonian engineered to mimic the p-adic metric causes the system to relax into a protected topological sector without active error correction, directly addressing the “thermodynamic wall.” This passive relaxation is an instance of autaxic **Feedback Dynamics** and **Resonance** guiding the system toward a coherent, low-entropy attractor.
 - **Ballistic Transport on the Bruhat-Tits Tree** (Quni-Gudzinis, 2026c) demonstrates that quantum walks on the tree exhibit ballistic transport $(O(D))$ time to traverse depth (D) , enabling efficient navigation of hierarchical latent spaces. This provides a geometric foundation for **explainable AI**, where the decision path of a model is a transparent walk on the tree, realising the autaxic **Principle of Interactive Complexity Maximization** in a computationally tractable form.
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3. Ratio-Based Valuation Theory: Scaling Ratios as Autaxic Parameters

3.1 q -adic Valuations and Base-Invariance

Ratio-based valuation theory (Quni-Gudzinis, 2026d) generalises classical p -adic valuations by replacing the prime (p) with an arbitrary positive **scaling ratio** $(q > 1)$. A (q) -adic absolute value $(|\cdot|_q)$ satisfies the same ultrametric inequality but is independent of any particular numeric base (decimal, binary, etc.). This base-invariance embodies the autaxic **Principle of Parsimony**: the fundamental generative mechanism should not depend on contingent representations.

3.2 Scaling Ratios as Operational Dynamics Parameters

Each scaling ratio (q) can be interpreted as a **parameter of an autaxic operational dynamic**. For example, (q) could set the branching factor of a critical transition, the strength of a feedback loop, or the resonance frequency that selects coherent patterns. The **Vladimirov operator** (D_q^α) , a pseudodifferential operator on (q) -adic spaces, generates diffusion whose spectral properties are determined solely by (q) —a concrete mathematical realisation of the autaxic idea that the “laws” emerge from the intrinsic parameters of the generative engine.